
ST-SIMDIFF: BALANCING SPATIOTEMPORAL SIMILARITY AND DIFFERENCE FOR EFFICIENT VIDEO UNDERSTANDING WITH MLLMs

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ABSTRACT

Multimodal Large Language Models (MLLMs) incur substantial computational overhead on long videos because video understanding requires large sequences of visual tokens. Existing efficiency methods mainly reduce this burden by pruning or merging tokens according to importance or similarity. Although effective, these strategies underemphasize an essential property of video: changes, transitions, and turning points often carry the events that determine the semantic trajectory of the sequence. They also typically lack a unified account of spatio-temporal token relationships. We introduce ST-SimDiff, a training-free framework built on the principle that similarity should identify redundancy whereas difference should preserve events. ST-SimDiff first constructs a spatio-temporal graph over visual tokens to model their local spatial and temporal associations within a single structure. It then performs parallel dual selection: 1) similarity-based representative token selection applies community detection to retain central tokens from redundant, stable regions; and 2) temporal difference-based event token selection identifies content-changing points and preserves tokens associated with dynamic transitions. The resulting token subset compactly represents both persistent visual content and event-defining changes. Extensive experiments demonstrate that ST-SimDiff consistently surpasses state-of-the-art token compression approaches while substantially reducing inference cost. Our code will be released upon acceptance.

1 INTRODUCTION

The rapid development of Large Language Models (LLMs) has accelerated the emergence of Large Vision-Language Models (LVLMs), which now exhibit strong image and video understanding capabilities (Zhang et al., 2024c; Liu et al., 2024c). In video settings, current LVLMs typically sample a video into frames and convert each frame into hundreds or even thousands of visual tokens before passing them to the LLM (Bai et al., 2025; Li et al., 2024). This paradigm is effective, but its token count grows rapidly with video duration and resolution, producing substantial computational and memory costs. Consequently, efficient video token processing has become central to enabling LVLMs in long-video analysis and real-time interaction scenarios (Fu et al., 2024b).

Existing efficiency methods for LVLMs primarily follow two lines of token reduction. Importance-based pruning removes visual tokens that receive low attention in later model layers, motivated by the observation that LVLM attention distributions contain substantial redundancy when processing visual inputs (Chen et al., 2024). Similarity-based merging or selection instead exploits the high feature similarity observed among visual tokens, either across adjacent video frames or within individual frames. For example, FrameFusion (Fu et al., 2024b) reduces redundancy by merging similar tokens from neighboring frames, whereas VisionZip (Yang et al., 2025) selects dominant tokens directly at the vision-encoder level.

Despite their success in reducing visual tokens, existing methods leave two limitations unresolved. First, many approaches model either spatial correlations within a frame or temporal correlations at corresponding positions across frames, rather than analyzing joint spatio-temporal structure. This separation restricts their ability to exploit complex redundancy patterns that arise from persistent

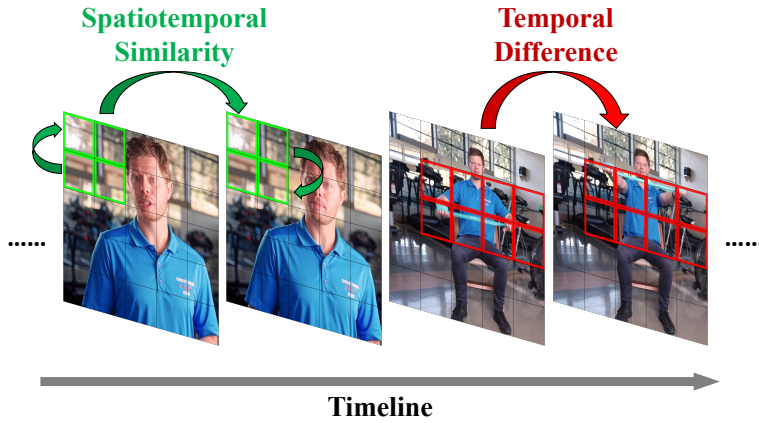


Figure 1: Core motivation of ST-SimDiff. Efficient video understanding requires simultaneous treatment of spatiotemporal similarity and temporal difference: similarity (left) supports compression of redundant information within frames and across adjacent frames, whereas difference (right) identifies key actions or events that define the video trajectory.

objects, backgrounds, and motions. Second, most methods optimize for informational commonality, such as importance or similarity, but do not explicitly protect visual difference. Because video semantics are often determined by transitions, sudden actions, and event boundaries, a compression rule that only favors common or representative content can discard precisely the tokens that explain what happened.

This paper advances a complementary view: similarity identifies redundancy, whereas difference identifies key moments. Under this view, effective video token compression should satisfy two coupled objectives: it should represent stable visual content with as few tokens as possible, while preserving the temporal changes that define events. We instantiate this principle as **ST-SimDiff**, a dual token selection framework built on a spatio-temporal graph. In the first component, **similarity-based representative token selection**, all visual tokens are treated as graph nodes connected according to spatio-temporal feature similarity. Community detection then identifies tightly connected token clusters corresponding to stable or persistent visual elements, and high-centrality tokens from each cluster are retained as representatives. In the second component, **difference-based event token selection**, temporal graph edges are examined to detect sharp drops in similarity between adjacent frames. These drops mark turning points, and the corresponding tokens are retained as event-critical evidence. The final visual input to the LLM is the union of representative tokens and event tokens. The resulting contributions are threefold:

- We formulate video token compression around the joint role of similarity and difference, establishing that redundancy removal and event preservation should be treated as complementary objectives in efficient VLM inference.
- We introduce a spatio-temporal graph framework that models spatial and temporal token relationships within a unified structure, together with a dual selection strategy for retaining both representative stable tokens and pivotal event tokens.
- We provide extensive benchmark evidence showing that ST-SimDiff substantially reduces the visual token budget while maintaining or improving video understanding performance across evaluated settings.

2 RELATED WORK

2.1 LARGE VISION LANGUAGE MODELS

The rapid advancement of Large Language Models (LLMs) (Touvron et al., 2023; Achiam et al., 2023; Ouyang et al., 2022) has significantly propelled progress in multimodal understanding, lead-

ing to the emergence of numerous prominent Large Vision Language Models (LVLMs) (Liu et al., 2024b; Bai et al., 2023). These LVLMs typically process images or video frames by converting them into visual tokens via pre-trained visual encoders, which are then fed into LLMs. Various alignment modules commonly connect the visual encoder and the LLM, including MLPs (Liu et al., 2024a;b) and Q-Formers (Li et al., 2023; Dai et al., 2023). This architecture enables LVLMs to exhibit exceptional capabilities in multimodal tasks (Xi et al., 2025; Holdaway et al., 2024; Zhang et al., 2025a). Particularly in video understanding, they show promise in processing longer, higher-resolution complex videos (Bai et al., 2025; Li et al., 2024; Zhang et al., 2024c; Liu et al., 2024c).

However, this powerful capability also introduces significant computational challenges, especially in video understanding scenarios. Due to the need to process continuous sequences of frames, the number of visual tokens grows exponentially, far exceeding that of static images, posing a severe challenge to the training and inference efficiency of LVLMs. Therefore, how to efficiently process massive video visual tokens and reduce computational overhead remains a critical issue in current LVLM research.

2.2 VISUAL TOKEN COMPRESSION

Driven by the substantial overhead of video processing, visual token compression has become a critical method for improving the efficiency of Large Visual Language Models (LVLMs). Existing methods primarily approach compression from two aspects: token importance and similarity.

Earlier methods primarily focused on importance-based token pruning, aiming to identify and remove visual tokens that contribute less to model performance. These methods typically utilize attention scores or other feature metrics to quantify token importance, such as FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b). Subsequent works (Shang et al., 2024; Ye et al., 2025) have started to further consider token similarity to more effectively reduce redundancy and preserve critical information. These methods recognize that even important tokens can exhibit high similarity, leading to informational redundancy. For instance, FrameFusion (Fu et al., 2024b) combines similarity-based merging with importance-based pruning. VisionZip (Yang et al., 2025) reduces redundancy by selecting dominant tokens and merging contextual information. VisPruner (Zhang et al., 2025b) leverages visual cues to remove redundant items.

Despite significant progress in visual token redundancy, existing work hardly addresses the intricate, global spatio-temporal similarity relationships between tokens. In video scenarios, visual information is redundant both spatially and temporally. Existing similarity-driven methods, like FrameFusion, mainly only on the temporal similarity between adjacent frames, failing to utilize complex spatio-temporal relevant information. This limits their ability to fully exploit inherent video redundancy, restricting performance in extreme token reduction scenarios.

3 PROBLEM DEFINITION

Given a video V and a text query Q , a standard Large Vision-Language Model (LVLM) first encodes the video into a full sequence of N visual tokens, $T_{\text{full}} = \{t_1, t_2, \dots, t_N\}$. This process is computationally demanding, as the self-attention mechanism’s complexity scales quadratically with the token count N , i.e., $O(N^2)$. Our task is therefore to design an efficient token selection function, $f(\cdot)$, that takes the full sequence T_{full} and a compression ratio r to produce a much smaller subset $T_{\text{sub}} = f(T_{\text{full}}, r)$. The objective is to maximize the LVLM’s downstream task performance using this subset, subject to the constraint that its size $|T_{\text{sub}}|$ is approximately $r \cdot N$. The core challenge is to design the selection function $f(\cdot)$ to retain the most critical information within the subset T_{sub} .

4 ST-SIMDIFF

4.1 METHOD OVERVIEW

As illustrated in Figure 2, the proposed ST-SimDiff method comprises two key components that operate in parallel on a spatio-temporal graph constructed from the video’s visual tokens: Similarity-based Representative Token Selection (SRTS) and Difference-based Event Token Selection (DETS). First, in the SRTS module, we identify and condense the video’s stable and redundant content. By

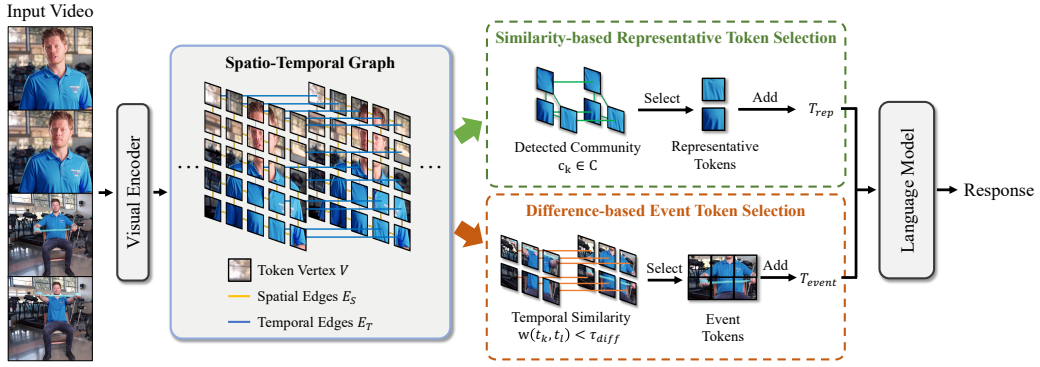


Figure 2: The overview framework of ST-SimDiff, which consists of three parts: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection, and Difference-based Event Token Selection.

applying community detection algorithms to the graph, we group highly similar tokens into clusters, which correspond to persistent visual elements like static backgrounds. From each cluster, we then select a few highly central tokens to form a representative set, T_{rep} . Second, the DETS module is designed to capture the video’s crucial dynamic shifts. It analyzes the temporal edges of the graph, identifying moments where the similarity between corresponding tokens in adjacent frames drops sharply. These points of high difference signify turning points, and the tokens framing these transitions are preserved as critical event tokens, forming the set T_{trans} . By synergistically combining these two components, ST-SimDiff generates a final token subset $T_{sub} = T_{rep} \cup T_{trans}$, which is both highly compact and information-rich, retaining stable content while precisely capturing key events.

4.2 SPATIO-TEMPORAL GRAPH CONSTRUCTION

Given a video, the vision encoder transforms it into a sequence of tokens $T = \{t_1, t_2, \dots, t_N\}$. Each token $t_k \in T$ corresponds to a specific spatio-temporal location and is associated with a feature vector $\mathbf{x}_k$. We define the location of each token t_k by its frame index $T(t_k)$, its spatial height index $H(t_k)$, and its spatial width index $W(t_k)$.

We model the relationships between these tokens using a spatio-temporal graph $G = (V, E)$, where the vertex set V represents all tokens in T . The edge set E is the union of two distinct subsets, $E = E_S \cup E_T$, which capture spatial and temporal relationships respectively. The spatial edges E_S connects tokens that are spatially adjacent within the same frame:

$$E_S = \{(v_i, v_j) \in V \times V \mid T(t_i) = T(t_j) \text{ and } |H(t_i) - H(t_j)| + |W(t_i) - W(t_j)| = 1\} \quad (1)$$

The temporal edges E_T connects tokens that are at the same spatial position but in adjacent frames, capturing temporal continuity:

$$E_T = \{(v_i, v_j) \in V \times V \mid H(t_i) = H(t_j), W(t_i) = W(t_j), \text{ and } |T(t_i) - T(t_j)| = 1\} \quad (2)$$

The weight of any edge $w(v_i, v_j) \in E$ is defined by the cosine similarity of the feature vectors of the corresponding tokens:

$$w(v_i, v_j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|} \quad (3)$$

This sparse graph structure efficiently encodes local spatial relationships and temporal continuity, providing a unified foundation for our dual-path token selection strategy.

4.3 SIMILARITY-BASED REPRESENTATIVE TOKEN SELECTION

Video content is inherently characterized by substantial spatio-temporal redundancy. For instance, in a static scene that persists for several seconds, the background tokens exhibit high similarity over

time. Similarly, for an object moving across the screen, its corresponding tokens, despite changing spatial positions, maintain a high degree of semantic correlation. Existing methods often address temporal and spatial similarities in isolation. In contrast, our spatio-temporal graph is designed to capture these joint similarities. Within this graph, semantically related tokens form dense “communities” or “clusters” based on their feature similarity, irrespective of their specific frame or location. Therefore, selecting representatives from each community emerges as a highly efficient strategy for compressing redundant information while preserving core semantics.

To accurately identify strongly correlated token clusters, we first threshold the graph G . We set a similarity threshold τ_{sim} and retain only the edges with weights above this threshold, forming a new graph $G' = (V, E')$, where $E' = \{(v_i, v_j) \in E \mid w(v_i, v_j) > \tau_{\text{sim}}\}$. Next, we apply a graph community detection algorithm (e.g., the Louvain method (Blondel et al., 2008)) on G' to identify token clusters $C = \{c_1, c_2, \dots, c_m\}$. For each community $c_k \in C$, we rank and filter its internal tokens based on their centrality. The centrality score $S_c(t_a)$ of a token $t_a \in c_k$ is defined as its average similarity with all other tokens within the community:

$$S_c(t_a) = \frac{1}{|c_k| - 1} \sum_{t_b \in c_k, b \neq a} w(t_a, t_b) \quad (4)$$

Subsequently, we set the intra-community retention rate to the globally preset compression ratio r , and preserve the top $\lceil |c_k| \cdot r \rceil$ tokens with the highest centrality scores from each community c_k . This uniform filtering process naturally handles all communities, including single-node communities composed of unique tokens (in which case the sole node is retained). Formally, the set of representative tokens, T_{rep} , is defined as:

$$T_{\text{rep}} = \bigcup_{k=1}^m \text{TopK}_{t \in c_k, \text{score}=S_c(t)}(\lceil |c_k| \cdot r \rceil) \quad (5)$$

where the TopK function returns the set of k elements with the highest scores from a given set. In summary, this strategy efficiently compresses the static and persistent content of the video by identifying and retaining the most central tokens within each semantic cluster.

4.4 DIFFERENCE-BASED EVENT TOKEN SELECTION

If similarity defines the “norm” of a video, then difference defines its “events”. A model focusing only on similarity may excel at understanding “what is” but struggle with “what happened”. A video’s plot is driven by key events, and the essence of an event is change—the appearance of a new object, the start of an action, or a scene transition. In our model, these changes manifest as a sharp difference in the features of temporally adjacent tokens. Therefore, accurately capturing these “turning points” of significant difference is crucial for understanding the video’s dynamic content and correctly answering questions like “when” and “why”.

Difference, particularly along the temporal dimension, often signals the occurrence of a key event. For example, the entry of a new object, a scene change, or the beginning/end of an action will cause drastic changes in the visual features at corresponding positions in adjacent frames. We specifically analyze the temporal edges (E_T) in our spatio-temporal graph. For any temporal edge $(v_k, v_l) \in E_T$ in the graph, which connects two temporally adjacent tokens t_k and t_l , we set a dynamic threshold τ (e.g., the 95th percentile of all temporal edge difference scores). When the difference score of a temporal edge (v_k, v_l) exceeds this threshold, i.e., $w(t_k, t_l) < \tau_{\text{diff}}$, we consider the subsequent token t_l as a critical event token and retain it. Formally, the set of event tokens, T_{event} , is defined as:

$$T_{\text{event}} = \{t_l \mid \exists t_k \text{ s.t. } (v_k, v_l) \in E_T, T(t_l) > T(t_k), \text{ and } w(v_k, v_l) < \tau_{\text{diff}}\} \quad (6)$$

With this strategy, tokens that signify moments of significant temporal change can be preserved. By capturing these key transitions, we ensure that the crucial dynamic aspects of the video’s narrative are not overlooked during the compression process.

4.5 OVERALL REDUCTION PROCESS

In the proposed framework, we first compute the representative token set T_{rep} and the event token set T_{event} in parallel, and take their union, $T_{\text{candidate}} = T_{\text{rep}} \cup T_{\text{event}}$, as the initial candidate set. To

precisely meet the target token count $N_{\text{target}} = \lceil r \cdot N \rceil$, we introduce a final pruning step. If the size of the candidate set, $|T_{\text{candidate}}|$, exceeds N_{target} , we remove the $|T_{\text{candidate}}| - N_{\text{target}}$ tokens with the lowest importance. Following existing work (Chen et al., 2024), the importance of a token is determined by its attention score in the shallow layers of the LLM. This step ensures that our method can flexibly meet any computational budget while preserving critical information.

In summary, our overall method adeptly combines two strategies: graph-based token selection and attention-based dynamic pruning. The former leverages the intrinsic structure of the video content (similarity and difference) to ensure the retention of core information, while the latter provides a flexible mechanism to precisely meet any given computational budget. This design makes our token compression process both principled and adaptive, thereby achieving a robust balance between efficiency and performance.

4.6 COMPUTATIONAL COMPLEXITY ANALYSIS

The computational overhead of our proposed ST-SimDiff framework can be analyzed in three main stages: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection (SRTS), and Difference-based Event Token Selection (DETS).

The initial stage involves building the graph. For each of the N visual tokens, we compute the cosine similarity with its spatially and temporally adjacent neighbors. As each token has a small, constant number of neighbors, this step requires a single pass over all tokens, resulting in a complexity of $O(Nd)$, where d is the feature dimension of each token.

Following graph construction, the SRTS module identifies token clusters. As detailed in our implementation (Section 5.2), we employ a connected components algorithm, which operates with a near-linear time complexity of $O(N + |E|)$. Since the number of edges $|E|$ is at most $3N$, this simplifies to $O(N)$. The subsequent filtering step within each community could be computationally intensive. A naive approach of calculating all-pairs similarity within each community c_k would have a complexity of $O(|C| \cdot |c_k|^2 \cdot d)$. To prevent this, we impose a constraint on the maximum size of any community. In the rare event that a community exceeds a predefined threshold (e.g., $|c_k| > \sqrt{N}$), we partition it. This ensures the filtering process remains efficient and does not dominate the overall complexity. While other mainstream graph clustering algorithms such as the Louvain and Leiden methods offer more complex community definitions, their computational complexity is higher, approaching $O(N \log N)$. Therefore, we opted for the simpler and highly efficient connected components algorithm to ensure minimal processing overhead.

The DETS module involves a single pass through all temporal edges in the graph to identify significant changes by comparing token similarities against a threshold. This process has a linear complexity of $O(Nd)$.

In summary, the total complexity of our ST-SimDiff framework is governed by these linear-time operations, culminating in an overall complexity of $O(Nd)$. This is substantially more efficient than the quadratic complexity of the self-attention mechanism, $O(N^2d)$, used in the Large Language Model. Therefore, our method provides significant token reduction with only a negligible computational footprint relative to the model’s main inference cost.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Baselines To comprehensively evaluate our method’s effectiveness, we use LLaVA-Video (Zhang et al., 2024c) and NVILA (Liu et al., 2024c) as our base models and compare our approach against a range of state-of-the-art video token compression techniques. These baselines cover diverse strategies, including **importance-based reduction methods** like FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b), and **hybrid reduction methods** like VisionZip (Yang et al., 2025), FrameFusion (Fu et al., 2024b), and PruMerge (Shang et al., 2024). To ensure a fair and complete comparison, we also include the original uncompressed model (Vanilla) as a performance upper bound and Random Sampling of an equivalent number of tokens as a lower bound.

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
LLaVA-Video	-	-	-	63.3	-	-	58.2	57.3
Token Retain Ratio $r = 30\%$								
FastV	68.2	58.6	51.2	59.3	48.0	59.9	53.5	51.3
PruMerge	70.2	57.3	52.2	59.9	49.6	60.5	54.7	50.9
FasterVLM	71.3	57.6	51.4	60.1	51.0	61.3	55.8	52.6
VisionZip	68.4	55.9	50.4	58.3	46.9	60.3	53.2	53.0
FrameFusion	74.0	59.8	50.0	61.3	49.7	63.3	56.0	53.0
ST-SimDiff (Ours)	74.7	61.9	53.0	63.2	52.5	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$								
FastV	73.9	61.4	51.3	62.2	50.1	62.2	55.7	54.7
PruMerge	72.3	58.7	52.8	61.3	51.4	63.2	56.9	54.6
FasterVLM	74.0	59.2	51.8	61.7	51.3	62.2	56.4	56.2
VisionZip	72.6	57.7	51.6	60.6	51.3	63.2	56.8	54.2
FrameFusion	74.6	61.2	52.0	62.6	51.7	64.2	57.6	55.8
ST-SimDiff (Ours)	76.2	62.4	52.7	63.8	52.3	64.3	57.9	57.3

Table 1: Performance comparison on long-form video understanding benchmarks on LLaVA-Video-7B for 64 input frames (%). The best performance among all methods is emphasized in **bold**.

Benchmarks To test our method’s capabilities in long-video understanding, we adopt three challenging benchmark datasets: VideoMME (Fu et al., 2024a), LongVideoBench (Wu et al., 2025), and EgoSchema (Mangalam et al., 2024). VideoMME is a comprehensive benchmark for foundational video understanding, featuring diverse video types and three length categories: Short, Medium, and Long. LongVideoBench evaluates the ability to retrieve and reason about details in long videos through a “referring reasoning” task, with videos of four lengths: 15s, 60s, 600s, and 3600s. EgoSchema is a question-answering dataset focused on long-term, first-person videos that tests the model’s ability to understand character intentions and causal chains of actions.

5.2 IMPLEMENTATION DETAILS

We use NVIDIA L20 GPUs with 48GB VRAM on an Ubuntu 22.04. The inference evaluation is conducted based on the *lmms-eval* (Zhang et al., 2024a) library. For LLaVA-Video, we follow its original setting of 64 input frames. To ensure a fair comparison of computational efficiency and resource usage under similar conditions, we establish a consistent setup by also setting the input frame count for NVILA to 64. As for the hyperparameters, we set the similarity threshold $\tau_{\text{sim}} = 0.8$, and the difference threshold $\tau_{\text{diff}} = 0.2$. Experiments are conducted with token compression rates of $r = 30\%$ and $r = 50\%$, respectively. In our practical implementation, considering the trade-off between time consumption and performance improvement, we opt to use connected component finding as community detection algorithm.

5.3 COMPARISON WITH STATE-OF-THE-ARTS

To comprehensively evaluate our proposed ST-SimDiff framework, we compare it against a range of mainstream efficient video language model (VLM) compression methods. All experiments are conducted on three widely-used long-form video understanding benchmarks: VideoMME, LongVideoBench, and EgoSchema. To validate the generalization ability and robustness of our method, we report detailed performance data on two different base models, LLaVA-Video-7B and

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
NVILA-Video	-	-	-	61.5	-	-	56.3	52.9
Token Retain Ratio $r = 30\%$								
FastV	69.4	55.3	48.8	57.9	50.8	55.6	53.0	49.7
PruMerge	71.0	53.7	49.9	58.2	49.3	58.1	53.4	47.5
FasterVLM	72.6	56.9	51.0	60.1	48.7	57.8	53.0	49.3
VisionZip	71.4	56.2	49.7	59.1	46.3	56.2	50.9	48.9
FrameFusion	72.1	55.7	48.7	58.8	51.3	59.3	54.9	51.3
ST-SimDiff (Ours)	73.3	57.7	49.7	60.2	51.7	59.2	55.2	51.7
Token Retain Ratio $r = 50\%$								
FastV	71.9	58.3	49.3	58.9	49.5	58.8	53.9	50.2
PruMerge	71.2	54.8	46.8	57.6	49.7	58.7	53.9	48.9
FasterVLM	74.6	57.8	50.1	60.8	48.5	58.1	53.0	50.5
VisionZip	72.8	58.1	50.6	60.5	50.4	58.9	54.4	50.3
FrameFusion	72.7	57.2	48.3	59.4	51.0	59.2	54.8	52.6
ST-SimDiff (Ours)	73.9	59.7	51.4	61.7	52.3	61.3	56.5	52.5

Table 2: Performance comparison on long-form video understanding benchmarks on NVILA-Video-8B for 64 input frames (%). The best performance among all methods is emphasized in **bold**.

NVILA-8B, under token retain ratios of 30% and 50%. The experimental results are presented in Table 1 and Table 2.

(1) ST-SimDiff consistently outperforms all competing methods across all test configurations. A particularly noteworthy finding is that at a 50% token retain ratio, the overall performance of our method on both base models not only surpasses other compression algorithms but, on some benchmarks, even matches or exceeds that of the original model using 100% of the tokens. This result validates the effectiveness and novelty of ST-SimDiff and prompts a deeper analysis of the strategies and limitations of existing baselines.

(2) In importance-based baseline methods (e.g., FastV, FasterVLM), FasterVLM typically exhibits relatively stronger performance. However, their common limitation lies in the inefficient handling of temporal redundancy prevalent in videos, as they tend to retain important yet repetitive tokens, thereby limiting information compression efficiency. Hybrid baseline methods (e.g., PruMerge, VisionZip, FrameFusion) demonstrate more competitive performance than importance-only approaches, with FrameFusion generally showing the most outstanding overall results. Nevertheless, the common optimization goal of these methods is still to identify and preserve representative “commonality” information, with a theoretical blind spot of potentially overlooking key narrative information driven by “turning points” and “abrupt events”.

(3) In summary, existing baselines reflects a deepening understanding of “redundancy” in videos, yet they still face two core challenges: first, a lack of synergistic analysis of spatio-temporal joint similarity between visual tokens, making it difficult to capture complex redundancies; and second, a general neglect of “difference” detection, failing to actively preserve key changes that define plot development. The performance gain of ST-SimDiff stems from addressing these issues by constructing a spatio-temporal graph to uniformly model complex spatio-temporal relationships and innovatively introducing difference detection.

Method	VideoMME	LongVideoBench	EgoSchema
Token Retain Ratio $r = 30\%$			
Baseline	60.3	56.2	54.8
+ Sim	Spatial	61.5	56.5
	Temporal	61.7	56.8
	Spa. + Tem.	62.6	57.0
++ Diff (Proposed)	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$			
Baseline	63.2	56.7	56.5
+ Sim	Spatial	63.3	57.0
	Temporal	63.7	57.2
	Spa. + Tem.	63.7	57.8
++ Diff (Proposed)	63.8	57.9	57.3

Table 3: Ablation results of different components (%).

5.4 ABLATION STUDY

To validate the effectiveness of each core component within our ST-SimDiff framework, we conducted a series of detailed ablation studies. We started with a Baseline strategy that includes only fundamental importance-based pruning and progressively introduced our proposed similarity selection module (+Sim) and difference selection module (++)Diff to quantify their respective contributions. All experiments were conducted at token retain ratios of 30% and 50%, with the results presented in Table 5.

(1) First, we evaluated the effectiveness of the +Sim module. The experimental results clearly indicate that introducing similarity modeling on top of the baseline consistently improves model performance. We further broke down the similarity strategy into three types: spatial-only (Spatial), temporal-only (Temporal), and spatio-temporal joint (Spa. + Tem.). The data shows that collaboratively modeling the spatio-temporal associations between visual tokens is the most effective way to compress redundancy.

(2) After integrating the optimal spatio-temporal joint similarity module, we introduced the difference selection module (++)Diff to form our complete framework. The addition of this module provides a significant performance leap, particularly under the high compression ratio of 30%. This demonstrates its vital complementary role: while the aggressive +Sim module focuses on redundancy, the ++Diff module acts as a safety net to identify and preserve unique event tokens that define the video’s narrative. In contrast, the module’s marginal gain is smaller at a 50% retention ratio. This is because the more lenient +Sim module is more likely to have already captured these event tokens, and the overall model performance is already approaching its upper bound. This analysis confirms our core motivation: the framework’s strength lies in its ability to balance similarity-based redundancy compression with the crucial preservation of difference-based events, especially in demanding high-compression scenarios.

5.5 COMPUTATIONAL COST ANALYSIS

To quantify the efficiency advantages of our proposed ST-SimDiff method in practical applications, we conducted detailed tests on its inference time and peak GPU memory usage under a 30% token budget. We compared it against the baseline model (Baseline LLaVA-Video) without any compression. The experimental results in Figure 3 cover video inputs lengths from 32 to 128 frames.

Inference Time ST-SimDiff demonstrates a significant speedup. As the number of input video frames increases, the average inference time of the baseline model grows sharply from 1.56 seconds (32 frames) to 6.50 seconds (128 frames). In contrast, our method effectively curtails this growth,

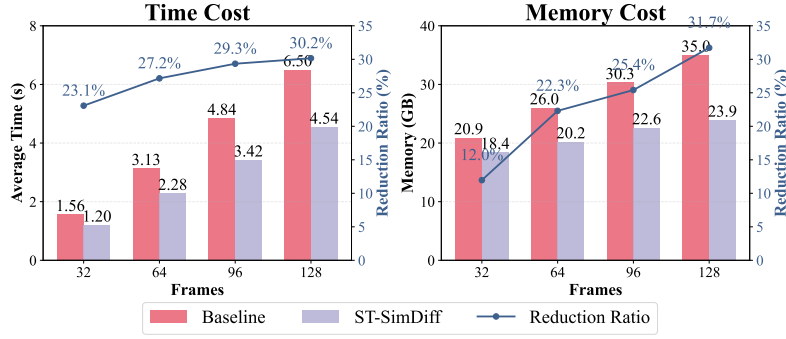


Figure 3: Computational cost comparison between our method and the baseline LLaVA-Video model in average inference time and peak GPU memory usage under a 30% token budget.

with an average time of only 4.54 seconds when processing 128 frames. This means that ST-SimDiff achieves an increasingly higher time-saving rate for longer videos, improving from 23.0% for 32 frames to 30.2% for 128 frames.

Memory Cost ST-SimDiff also performs exceptionally well. The peak GPU memory usage of the baseline model climbs linearly with video length, from 20.9 GB (32 frames) to 35.0 GB (128 frames). Our method significantly reduces memory pressure by substantially decreasing the number of tokens fed into the large language model. When processing 128-frame videos, ST-SimDiff’s peak memory usage is only 23.9 GB, saving 31.7% of the memory space compared to the baseline. This advantage enables our model to process longer videos on hardware with limited memory, greatly expanding its potential applications.

In summary, ST-SimDiff not only achieves leading performance but also demonstrates significant advantages in computational efficiency (both time and memory), proving its efficiency for video understanding.

6 CONCLUSION

In this paper, we address the computational inefficiency of Multimodal Large Language Models when processing long videos, a problem rooted in the excessive number of visual tokens and the neglect of critical content changes. We propose ST-SimDiff, a framework requiring no training that builds a spatiotemporal graph to model complex associations between tokens. The method utilizes a dual selection strategy where similarity based selection identifies redundancy via community detection and temporal difference based selection preserves tokens at key transition points. Extensive experiments on VideoMME and LongVideoBench show that ST-SimDiff significantly reduces processing time and memory usage while consistently surpassing state of the art performance. This work introduces a new research direction by balancing content representativeness with transitional events to create a more efficient representation of video data.

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A VISUALIZATION OF THE TOKEN SELECTION PROCESS

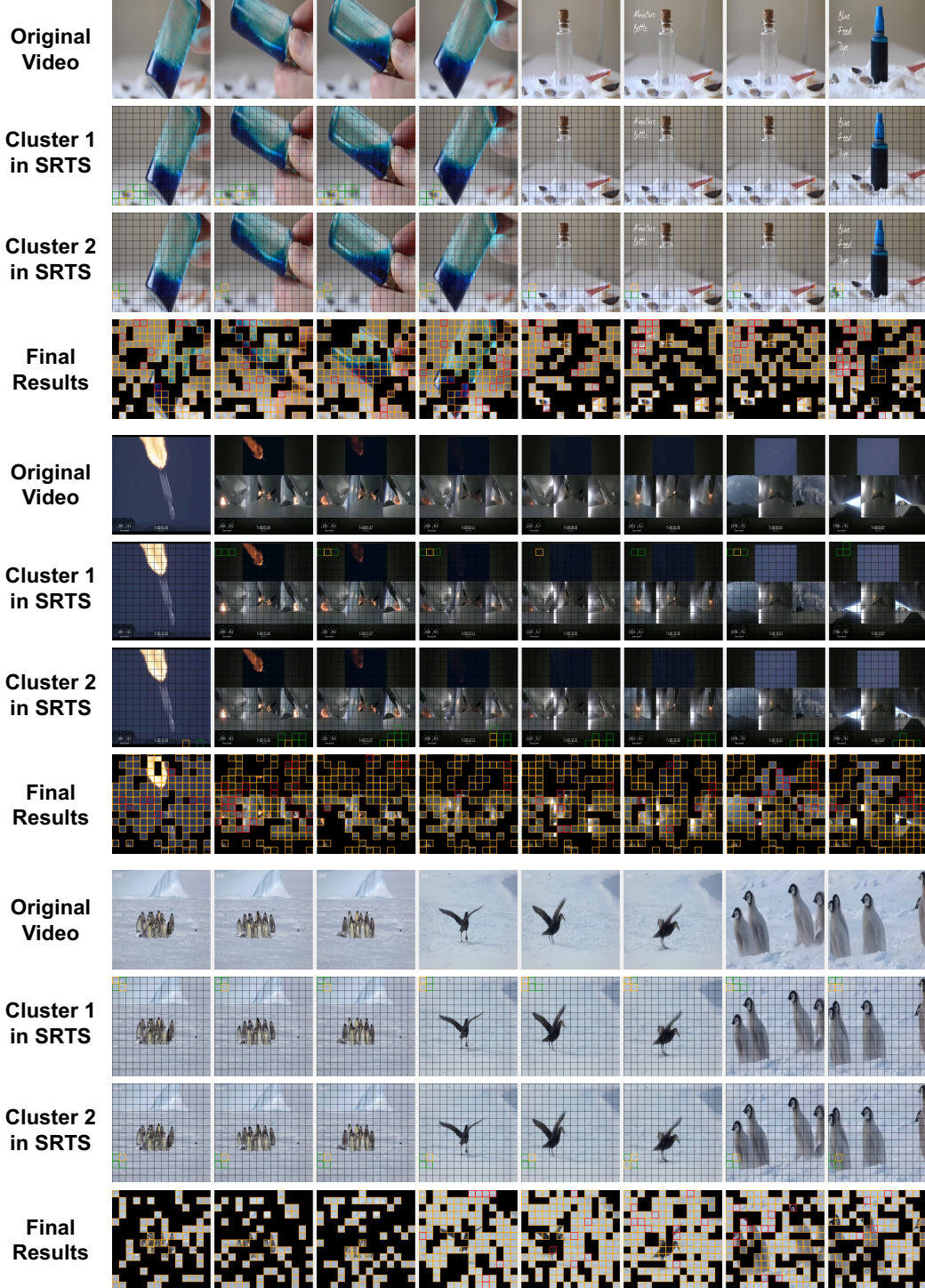


Figure 4: Visualization of the ST-SimDiff process. The visualization breaks down the token selection into communities detected by SRTS (shown as grids in Cluster 1 and 2) and event-driven tokens detected by DETS. The final result highlights the synergy between sparse representative tokens (yellow) for stable content and dense event tokens (red) for dynamic actions.

To provide a more intuitive understanding of the ST-SimDiff framework, we present some visualization samples of the intermediate and final token selection results in Figure 4. The figure illustrates the process on a sample video sequence featuring a dynamic object manipulation task. First, regarding the Similarity-based Representative Token Selection (SRTS), the rows labeled “Cluster 1” and “Cluster 2” demonstrate how our graph community detection algorithm functions. It successfully groups spatially and temporally redundant regions, such as the static background and stationary objects, into cohesive communities. This visualization confirms that SRTS effectively identifies stable visual elements and compresses them by selecting only a few central “representative tokens,” which appear as sparse yellow boxes in the final results. This mechanism allows the model to handle massive background redundancy efficiently.

Complementing this, the Difference-based Event Token Selection (DETS) plays a critical role during dynamic moments. As seen in the “Final Results” row, tokens triggered by high temporal difference are explicitly marked with dense red bounding boxes. These red boxes align perfectly with the moving hand and the shifting blue liquid, verifying that our method precisely locates “turning points” where the temporal difference exceeds the threshold (τ_{diff}). This ensures that fine-grained action details and rapid visual changes are preserved, allowing the model to capture the essence of the event without being overwhelmed by static data.

Finally, the synergistic effect of these two components is evident in the overlay of the final results. The visualization demonstrates a powerful balance: the sparse yellow tokens represent the stable background, while the dense red tokens capture the rapid motion. This contrast provides direct visual evidence of our core motivation. ST-SimDiff avoids the blind spots of similarity-only approaches by densely sampling during rapid changes, while simultaneously pruning the spatiotemporal redundancies that importance-based methods often miss. Consequently, our framework achieves an optimal balance between computational efficiency and the preservation of critical visual information.

B COMPARISON RESULTS ON QWEN2.5-VL

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
Qwen2.5-VL-64	74.2	62.7	51.8	62.9	54.2	64.8	59.2	63.0
Qwen2.5-VL-128	77.4	65.8	56.1	66.4	56.2	65.1	60.4	63.9
Token Retain Ratio $r = 30\%$								
ST-SimDiff-64	73.8	61.7	51.8	62.4	52.2	62.1	56.8	62.7
ST-SimDiff-128	75.4	64.7	55.2	65.1	53.1	65.0	58.6	64.4
Token Retain Ratio $r = 50\%$								
ST-SimDiff-64	74.2	62.1	52.2	62.9	54.2	63.2	58.4	62.7
ST-SimDiff-128	76.8	66.3	55.8	66.3	54.9	65.3	59.8	64.5

Table 4: Performance comparison on long-form video understanding benchmarks on Qwen2.5-VL-7B for 64 and 128 input frames (%).

To further validate the generalization capability of our proposed framework, we conducted additional experiments on the recently released Qwen2.5-VL model. These evaluations were performed on the VideoMME and LongVideoBench benchmarks to confirm that our method’s efficacy is not limited to a specific model architecture but can be broadly applied.

The empirical results, presented in Table 4, demonstrate the strong performance of ST-SimDiff on the Qwen2.5-VL model. At a token retention ratio of 30%, our method achieved scores of 62.4 on VideoMME and 56.8 on LongVideoBench. When the token budget was increased to a 50% retention ratio, the performance improved to 62.9 on VideoMME and 58.4 on LongVideoBench. It is particularly noteworthy that at the 50% ratio, the performance on VideoMME matches the upper-bound performance achieved using all tokens, and the score on LongVideoBench is highly comparable to its full-performance counterpart. These findings affirm the excellent generalization ability of our framework.

C ABLATION STUDY RESULTS ON NVILA

Method	VideoMME	LongVideoBench	EgoSchema
Token Retain Ratio $r = 30\%$			
Baseline	59.3	53.8	50.2
+ Sim	Spatial	59.5	54.5
	Temporal	59.6	54.3
	Spa. + Tem.	59.9	54.6
++ Diff (Proposed)	60.2	55.2	51.7
Token Retain Ratio $r = 50\%$			
Baseline	60.9	55.3	51.5
+ Sim	Spatial	61.1	55.7
	Temporal	61.3	55.6
	Spa. + Tem.	61.4	56.0
++ Diff (Proposed)	61.7	56.5	52.5

Table 5: Ablation results of different components on NVILA-Video-8B model (%).

To further validate the generalization ability of each component in our framework across different base models, we also conducted a series of detailed ablation studies on the NVILA-Video-8B model (Liu et al., 2024c). The experimental setup is consistent with the ablation study in the main paper; we start with a Baseline strategy that includes only fundamental importance-based pruning and progressively introduce the similarity selection module (+ Sim) and the difference selection module (++ Diff).

As shown in Table 5, the experimental results once again validate the effectiveness of our framework design, and the trends are highly consistent with the performance on LLaVA-Video-7B. First, after introducing the similarity module (+ Sim) on top of the baseline, the model’s performance shows a steady improvement at both 30% and 50% token retain ratios. Within the similarity module, we compared three strategies: spatial-only (Spatial), temporal-only (Temporal), and spatio-temporal joint (Spa. + Tem.). The results show that spatio-temporal joint modeling (Spa. + Tem.) is the most effective. For example, with $r = 50\%$, the spatio-temporal joint strategy achieved a score of 61.4% on VideoMME, outperforming the spatial-only and temporal-only strategies. This again demonstrates the importance of collaboratively considering spatio-temporal associations for effectively identifying redundant information. After integrating the optimal spatio-temporal joint similarity module, we further introduced the dissimilarity selection module (++ Diff) to form our complete ST-SimDiff framework. The table clearly shows that the addition of this module brought the most significant performance gains. This result strongly demonstrates the effectiveness and generalization ability of the two core modules in our framework (similarity and dissimilarity selection) and once again validates our core idea of simultaneously handling redundancy and change.

D ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON LLaVA-VIDEO

In our framework, the similarity threshold τ_{sim} and the difference threshold τ_{diff} are two key hyperparameters that respectively influence the selection of representative and transitional tokens. To investigate their impact, we conducted a series of ablation studies, with the results shown in Figure 6. The experiments show that the impact of both parameters on model performance follows a similar trend, first rising and then falling, while demonstrating good robustness within a certain range. For τ_{sim} , a value that is too low leads to imprecise community detection, while a value that is too high can disrupt the integrity of semantic clusters. For τ_{diff} , a value that is too low can introduce noise due to over-sensitivity, whereas a value that is too high may cause the model to miss key events. According to the experimental results, the model achieves optimal performance at $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$. Therefore, we adopted this optimal configuration for all other experiments.

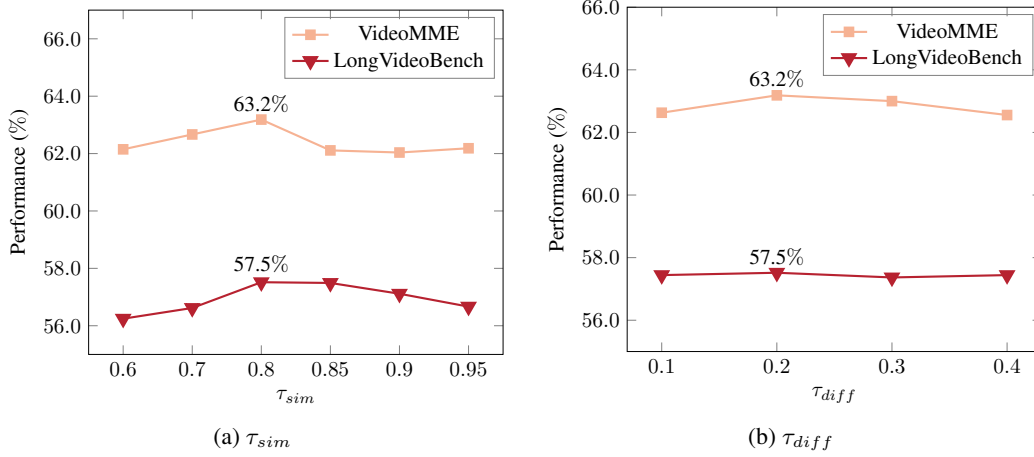


Figure 5: Ablation study results for different values of τ_{diff} and τ_{sim} on VideoMME and LongVideoBench.

E ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON NVILA-VIDEO

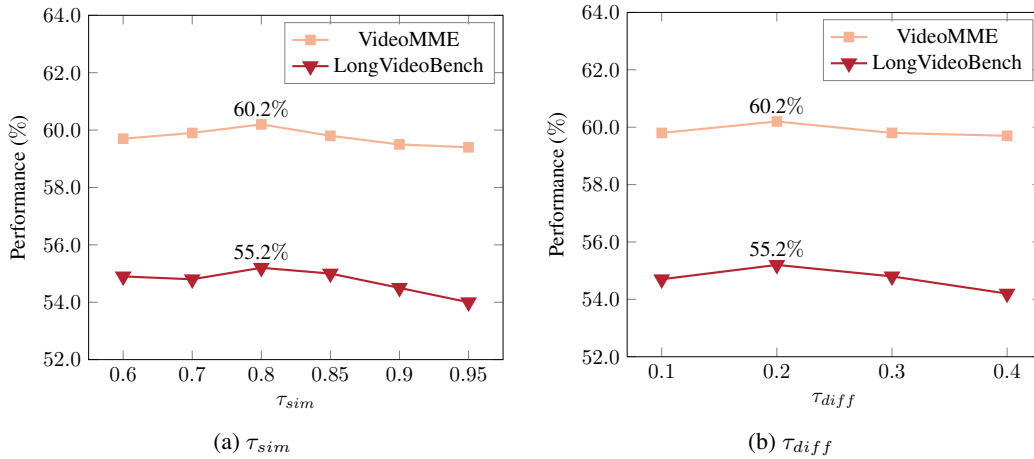


Figure 6: Ablation study results for different values of τ_{diff} and τ_{sim} on NVILA-Video-8B.

To determine the robustness of the key hyperparameters in our method, we conducted a series of ablation studies on the NVILA-Video-8B model. As shown in Fig. 6, we evaluated the impact of the similarity threshold, τ_{sim} , and the difference threshold, τ_{diff} , on the model’s performance across two benchmarks: VideoMME and LongVideoBench.

Impact of Similarity Threshold τ_{sim} . We tested different values for τ_{sim} within the range of $[0.6, 0.95]$. The experimental results show that the model’s performance is relatively insensitive to changes in this threshold. On the VideoMME dataset, peak performance of 60.2% was achieved at $\tau_{sim} = 0.8$. Similarly, on the LongVideoBench dataset, performance was also optimal at $\tau_{sim} = 0.8$, reaching 55.2%. While performance slightly decreased when the threshold was too high or too low, the overall fluctuation was minimal, confirming that our method maintains good stability across a wide range of similarity thresholds.

Impact of Difference Threshold τ_{diff} . We tested τ_{diff} in the range of $[0.1, 0.4]$. Similar to τ_{sim} , the model’s performance demonstrated strong robustness to variations in τ_{diff} . In both benchmarks, the optimal performance was achieved around $\tau_{diff} = 0.2$, with VideoMME at 60.2% and LongVideoBench at 55.2%. This indicates that setting the difference threshold around 0.2 is most

effective for capturing key events in the video, thereby maximizing model performance while compressing tokens.

In summary, these ablation studies confirm the robustness of our proposed method with respect to its key hyperparameters. The results validate our choice of using $\tau_{\text{sim}} = 0.8$ and $\tau_{\text{diff}} = 0.2$ in our main experiments, as these values consistently yield optimal performance across different benchmarks and different base models.